

SHAURYA SHARTHAK

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML engineer with hands-on production experience building **hybrid RAG systems**, **training infrastructure for 7B–14B parameter LLMs**, and **RLHF/GRPO pipelines**. Published first-author research on tokenizer transplantation (4 citations). Founded Tinycompany-AI with **155+ models** and **54 datasets (25M+ rows)** on HuggingFace. Engineered open-source PyTorch/XLA TPU training stack (**234 GitHub stars**) reducing HBM usage by 35%. Experienced in end-to-end ML pipelines from data curation to deployment.

EXPERIENCE

AI Engineer (Intern)

08/2024 – 01/2026

ResoluteAI Software Inc.

Bangalore, India

- **Hybrid RAG System:** Architected a multi-signal retrieval pipeline (TF-IDF + BM25 + Neo4j GraphDB + Redis KV cache + vector similarity) improving retrieval hit-rate by **45%**.
- **Pipeline Optimization:** Refactored inference pipelines, achieving **Rs.25.8 Lakhs/year** (~\$31K) cost savings and **30% latency reduction**.
- **AI Assistant Automation:** Built Pandas agent-based ticket automation system, gaining **30% operational efficiency** and **20% faster response times**.
- **Engineering Practices:** Agile sprints, Git feature branches, Postman API testing, CI/CD with Jenkins.

Data Engineer

2025

TinyCompany AI

Remote

- Curated **30K-entry Hindi instruct dataset** and developed **HINGLISH-LIMA** for code-mixed LLM training.
- Contributed to NLP domain expertise, red-teaming, and RLHF annotation workflows.

TECHNICAL SKILLS

ML Frameworks: PyTorch, PyTorch/XLA, TensorFlow, HuggingFace (Transformers/Datasets/PEFT/TRL), Scikit-learn

Training Infra: TPU v2-8, Distributed Training, GSPMD Model Parallelism, Flash Attention, Unsloth, LoRA/QLoRA

NLP & LLMs: Tokenizer Design (BPE/SentencePiece), RLHF/GRPO, RAG (BM25/TF-IDF/Vector), LangChain

Languages: Python (Expert), SQL, C++, Bash

Databases: ChromaDB, Pinecone (Vector) — Neo4j (Graph) — Redis (KV) — PostgreSQL

DevOps: Git/GitHub, Docker, Jenkins CI/CD, AWS (EC2/S3), REST APIs, Postman

PUBLICATIONS

Achieving Tokenizer Flexibility in LLMs via Heuristic Adaptation and Supertoken Learning

arXiv:2505.09738 | 2025

- **Lead Author** (S. Sharthak, V. Pahalwan, A. Kamath, A. Shirawalmath) | **4 citations** in 6 months
- Proposed **TokenAdapt**: model-agnostic tokenizer transplantation achieving **2× improvement** in zero-shot perplexity vs. ReTok and TransTokenizer on Llama-3.2-3B and Qwen2.5-3B.

KEY PROJECTS

PyTorch/XLA TPU Training Infrastructure | 234 stars, 27 forks

2024

- Engineered end-to-end training stack for **TPU v2-8** enabling free fine-tuning of **7B–14B** models; resolved PyTorch eager-execution vs. XLA graph-compilation bottlenecks for Llama and Mistral architectures.
- Built **Optimized-Attention-Torch-XLA** and custom trainer with **GSPMD model parallelism**, reducing HBM memory footprint by **35%** vs. standard XLA implementations.

GRPO/RLHF Training & Large-Scale Dataset Engineering

2024 – 2025

- Trained **Qwen2.5-GRPO-7B** using Group Relative Policy Optimization; curated **3.5M+** math samples and **51K** clarification-seeking conversations (**Be-More-Specific-USER**, 9 likes).
- Built **54 datasets (25M+ rows, ~140 GB)** spanning medical, math, code, reasoning, DPO/RLHF, and Indic NLP; trained **155+ models** deployed on HuggingFace Hub.

QTK-81K Multilingual Tokenizer | HuggingFace

2025

- Designed 81K vocabulary BPE tokenizer achieving **4× better Hindi tokenization** (10K vs. 40K tokens) compared to Phi-4; listed on HF multilingual tokenizer leaderboard.

Cybersecurity Penetration Testing

2024

- Performed vulnerability assessments for **3+ MNC web applications** using Burp Suite; improved security posture for systems processing **2M+ transactions**.

EDUCATION

Chhattisgarh Swami Vivekananda Technical University

2022 – 2026

B.Tech (Hons.) Computer Science (AI/ML)

Bhilai, India

Jawahar Navodaya Vidyalaya

2018 – 2021

Senior Secondary (GPA: 3.86/4.0) — Secondary (GPA: 3.6/4.0)

Kurukshetra, India